

Scalability & Future Plan

Date	30 September 2026	Team ID	6ab22029e880316a482c5f2a
Project Name	EduGenie – AI-Powered Educational Assistant	GitHub	https://github.com/theopintheo/EduGenie
Team Members	Theopin B, Antony Ragul A, Saravanan S, Karthick M	Status	Completed / working implementation

Current System Limitations

No.	Limitation	Impact	Priority
1	No persistent learner history	Progress cannot be stored across sessions.	High
2	No production authentication/rate limiting	Public deployment needs protection.	High
3	No measured load-test evidence	Capacity is not benchmarked.	Medium
4	No multilingual/PDF input	Limits formats and reach.	Medium

Scalability Plan

Aspect	Current State	Proposed Upgrade
User Load	Single FastAPI app	Managed hosting/reverse proxy and horizontal scaling.
Data Storage	No persistent DB	PostgreSQL/MongoDB for users, history and progress.
Performance	Synchronous AI calls	Caching, async jobs/queues, model fallback and observability.
Security	Environment key management	Auth, authorization, rate limiting, audit logs and secret management.

Future Roadmap

Phase	Enhancement	Expected Impact
2	Authentication, history, progress dashboard	Personalized continuity
3	Multilingual output, PDF/image input, accessibility	Broader usability
4	Adaptive learning, teacher/parent dashboard, group study	Advanced personalization